

SUMMARY

Test Automation Engineer with 3 years of experience, currently deployed on a German automotive OEM (MBRDI) via Knodtec Solutions. System-level testing for FOTA functionality involving proprietary TCU, achieving up to 99% test coverage across supported functionalities. Hands-on experience with Ethernet, CAN, and Flexray protocols, UDS protocol, DoIP flashing, and OTA update validation.

SKILLS

- Embedded Systems, C, C++
 - Automotive Testing
 - OTA/FOTA/Remote Update
 - Telematics Control Unit
 - DoIP Flashing
 - UDS Protocol
- AUTOSAR
 - Python, CAPL, C, C++
 - Ethernet, CAN, Flexray
 - Git & Github
 - UART, I2C, SPI
 - Vector Stack
- CAPL Scripting
 - Software Testing
 - JavaScript/Typescript
 - LVGL library
 - SQL
 - Agile Methodology

WORK EXPERIENCE

Knodtec Solutions Pvt. Lt.

Test Automation Engineer Jan 2024 - Present

- Developed a Python-based test automation framework, automating 200+ testcases and reducing manual regression testing time by over 40%.
- Setup and debugged HIL-testing environment from scratch using Vector Automotive Software Stack (CANoe).
- Working in close proximity with hardware team on the Bench Setup, interfacing multiple ECUs such as telematics control unit, Ignition ECU (EIS), Head unit, etc.
- Automated End-to-End OTA update flow to perform release acceptance testing. Designed intuitive and robust GUI for tracking live testing results and progress.
- Ensured 100% automation coverage by leveraging CI/CD pipelines on Jenkins for the project.
- Performed hardware log analysis, to analyze defect bugs across different regions globally.
- Performed frequent DoIP flashing for software build analysis and OTA flashing, emulating real-life scenarios via in-house backend solutions.
- Conducted internal state machine flow analysis to support testcase development and test environment management.
- Achieved up to 99% test coverage across supported functionalities through routine functionality enhancements.
- Investigated high-priority bug tickets to resolve production line failures.
- Automated Vector CANoe simulations and CAPL script execution via remote calls from Python processes.
- Collaborated cross-functionally to expand testing infrastructure and ensure timely test maturity report submission.
- Conducted software qualification testing and system integration testing, generating structured bug reports from test results.

Software Engineering Intern July 2023 - Jan 2024

- Hands-on working experience with STM32 microprocessor.
- Developed Human machine interface (HMI) using LVGL library in C.
- Designed and developed responsive embedded GUI screens on STM32 using LVGL, including dashboards, settings pages, pop-ups, animations, and touch-driven navigation flows.
- Optimized LVGL rendering performance and memory usage on resource-constrained STM32 hardware by tuning frame buffers, display refresh logic, and widget updates for smooth UI response.
- Worked on firmware for Renesas RX23W and Onsemi RSL10 with modules BLE, GPIO, Timers (GPT, RTC, etc.), Low power modes, FOTA.
- Used pUTTY terminal for debugging.
- Comprehensive experience of Embedded C coding and flashing build files on a custom boards.

EDUCATION

B. Tech. in Electronics and Communications Engineering 2019 - 2023

Government Engineering College, Ajmer

- Graduated the college with an overall 8.5 CGPA.